

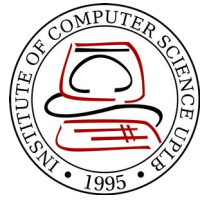
CMSC 137

Data Communications and Networking

Lecture Module
09 – Bandwidth Utilization



University of the Philippines
LOS BAÑOS



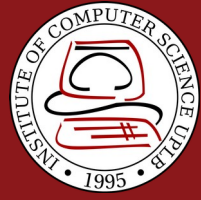
Module 9: Bandwidth Utilization

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Lecturer

Introduction



- Bandwidth utilization is the wise use of available bandwidth to achieve specific goals
- Efficiency can be achieved by multiplexing
- Privacy and anti-jamming can be achieved by spread spectrum

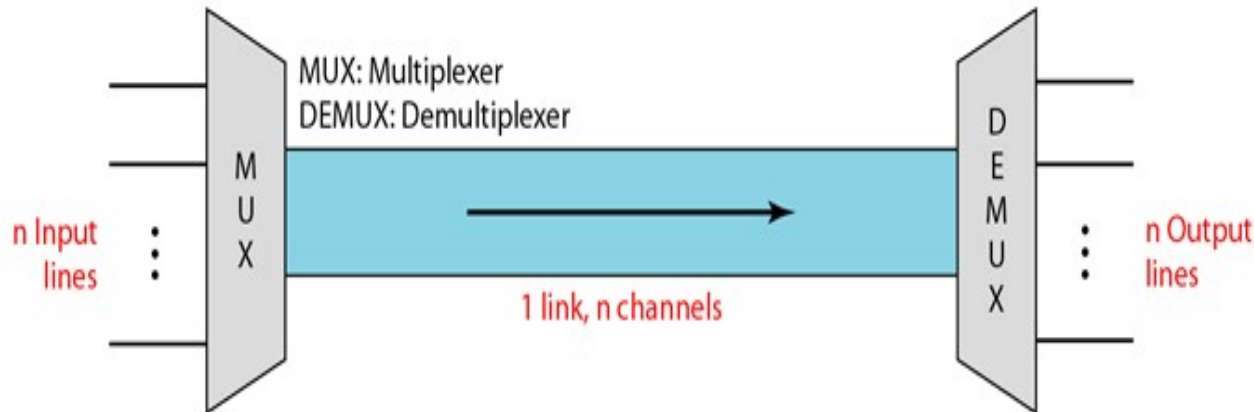


Multiplexing

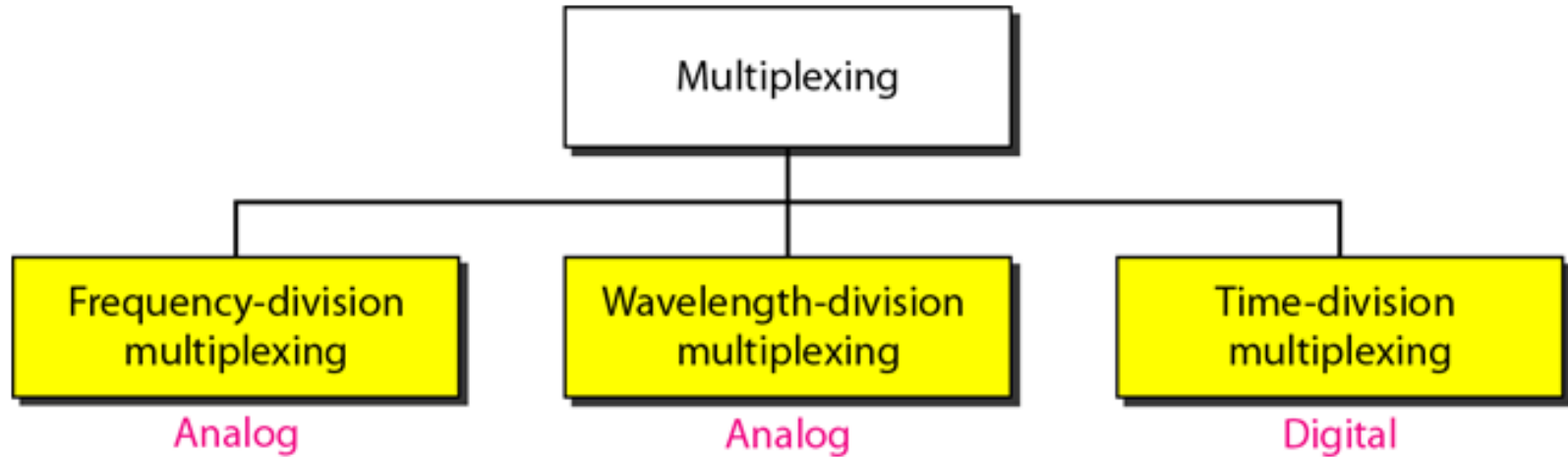
Multiplexing



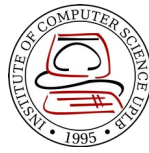
- Set of techniques that allow simultaneous transmission of multiple signals across a single link
- Bandwidth of medium is greater than bandwidth needs of connected devices, link can be shared
- **Channel** – portion of a link that carries a transmission between a given pair of lines



Categories of Multiplexing



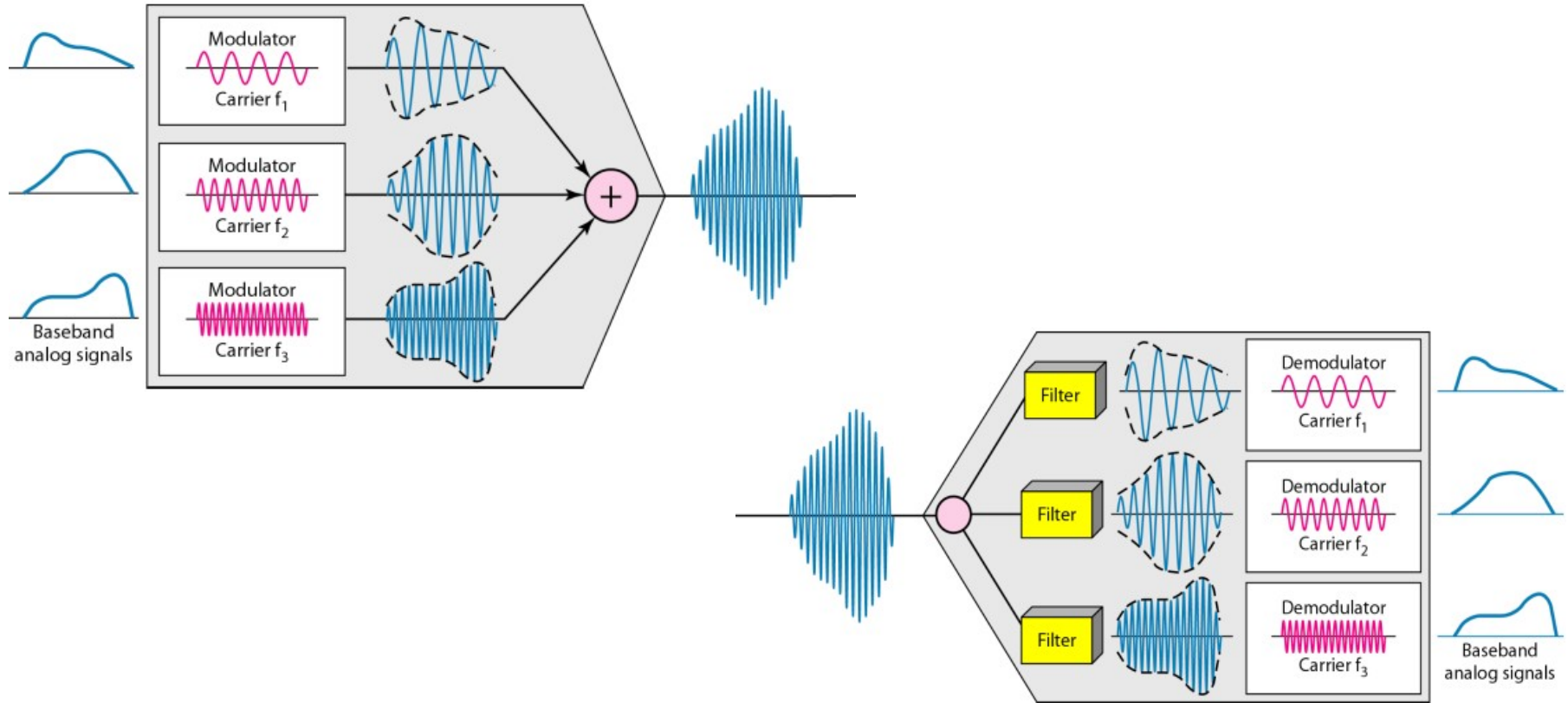
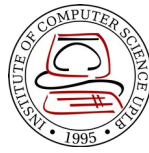
Frequency Division Multiplexing (FDM)



- Analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signal to be transmitted
- Modulated signals are combined into a single composite signal
- Channels can be separated by **guard bands** (strips of unused bandwidth) to prevent signals from overlapping



FDM Process

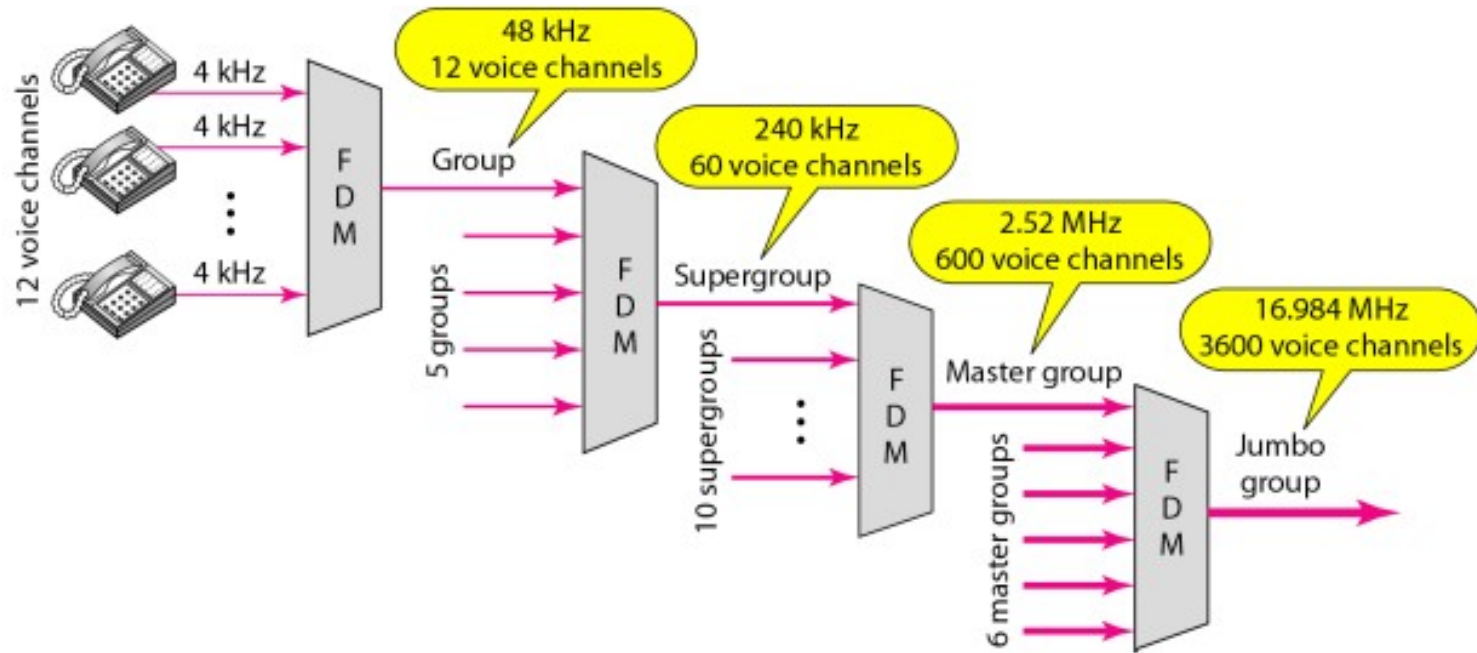
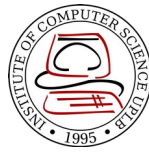


Analog Carrier System

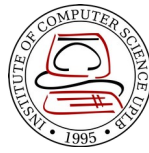


- Hierarchical system used by telephone companies that is made up of *groups*, *supergroups*, *master groups* and *jumbo groups*
- 12 voice channels are multiplexed onto a higher bandwidth line to create a group. A group has 48 kHz of bandwidth and support 12 voice channels
- At next level, up to 5 groups can be multiplexed to create a composite signal called a supergroup. A supergroup has a 240 kHz and support up to 60 voice channel

Analog Carrier System

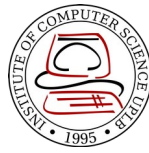


Other applications of FDM

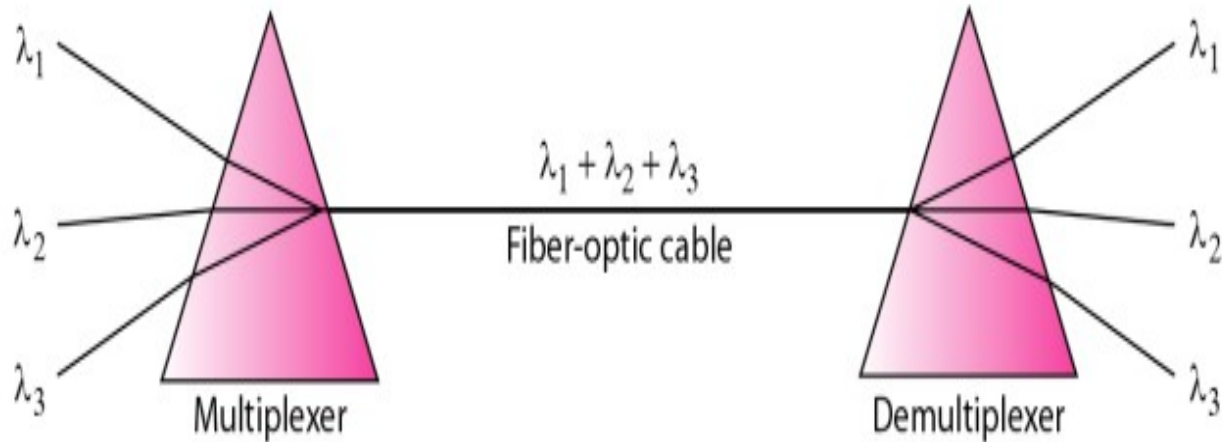


- AM and FM radio broadcasting
 - 530 to 1700 kHz assigned to AM radio
 - 88 to 108 MHz assigned to FM radio (each station needs 200 kHz)
- Television broadcasting (TV channel has 6 MHz each)
- First generation of cellular telephones
 - Each user is assigned two 30 kHz channels

Wavelength Division Multiplexing (WDM)



Analog multiplexing technique that combines optic signals

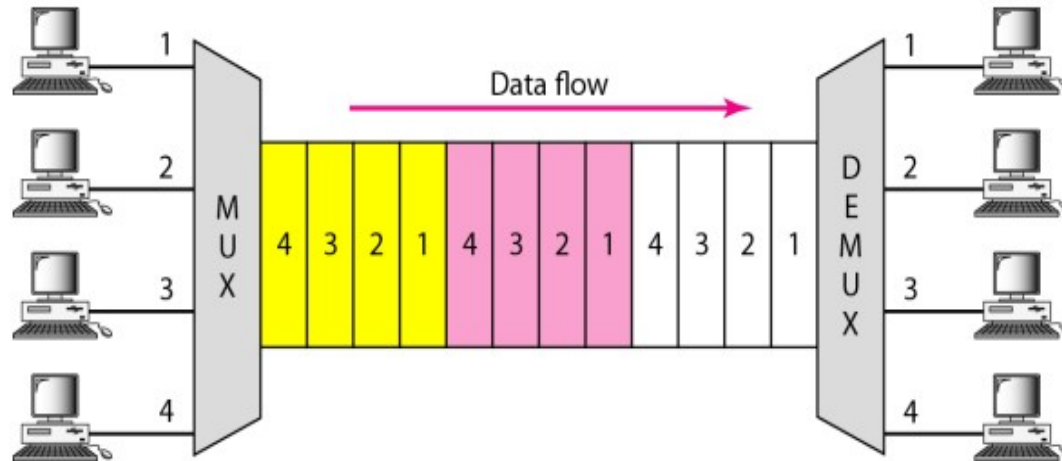


- One application is the SONET network
- New method, called **dense WDM** (DWDM) can multiplex a very large number of channels by spacing channels very close to one another (greater efficiency)

Time-Division Multiplexing (TDM)



- Digital multiplexing technique for combining several low-rate digital channels into one high-rate channel

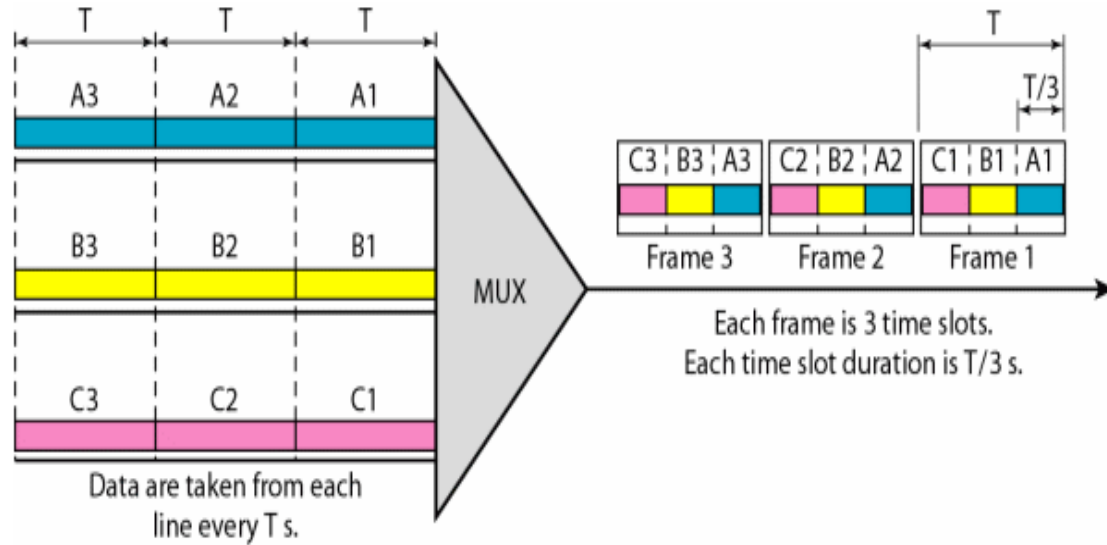


- 2 different schemes: [synchronous TDM](#) and [statistical TDM](#)

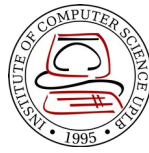
Synchronous TDM



- Each input connection has an allotment in the output even if it is not sending data

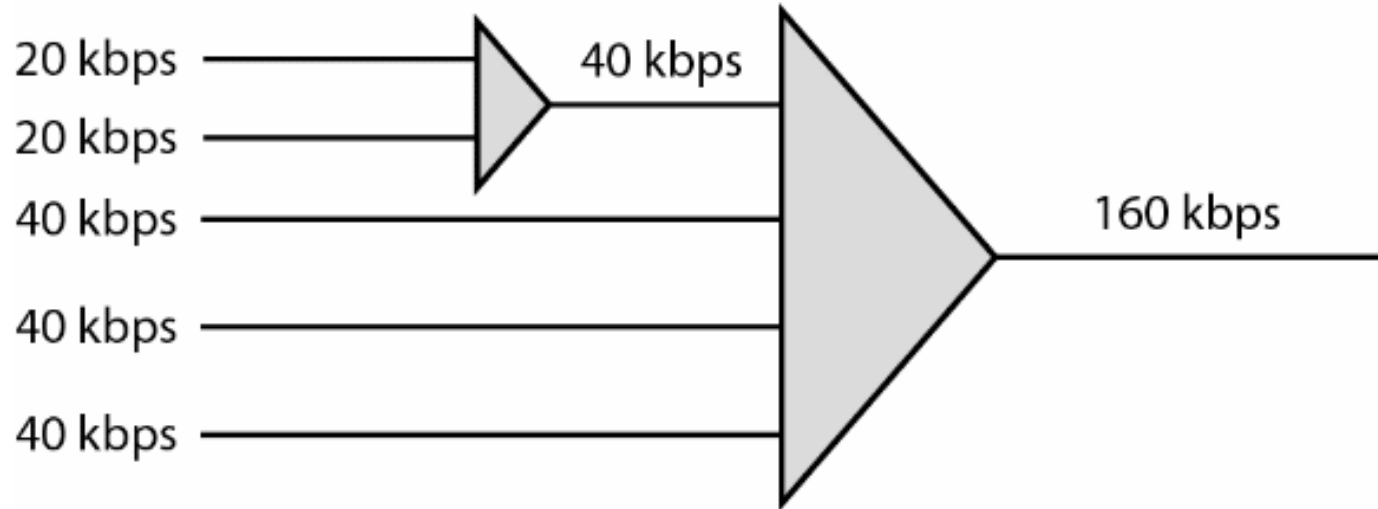


Synchronous TDM

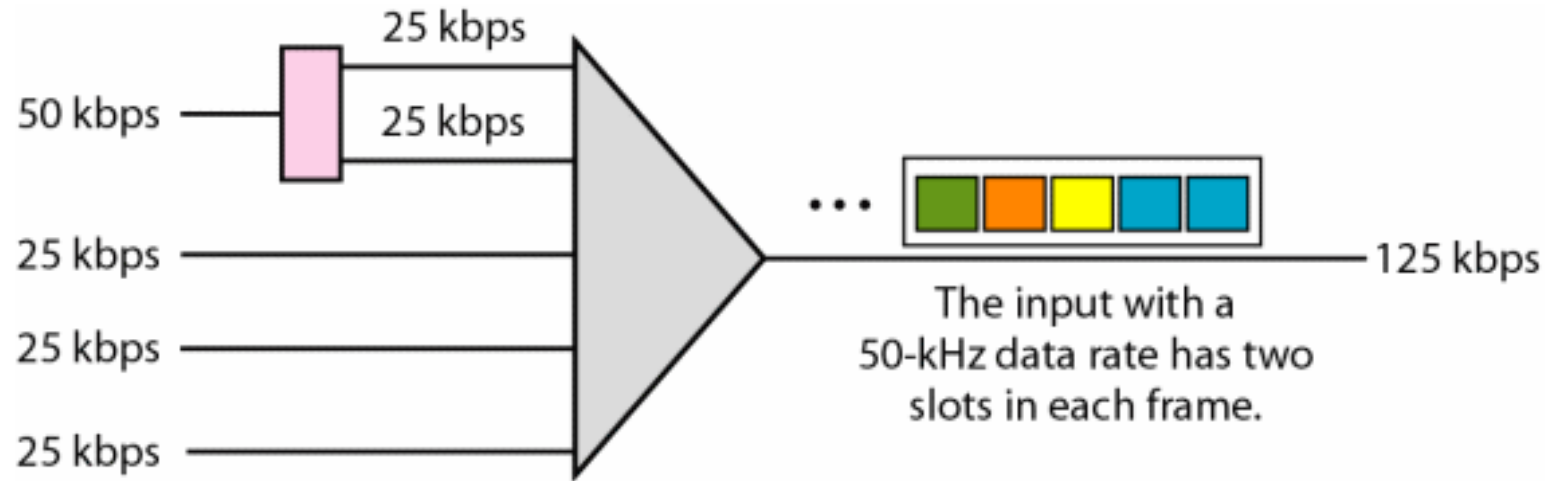
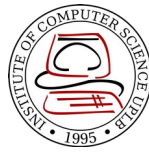


- In synchronous TDM, the data rate of the link is n times faster, and the unit duration is n times shorter
- What if the data rates of the inputs are different?
 - **Multilevel multiplexing** – data rate of input is a multiple of others inputs
 - **Multiple-slot allocation** – more than one slot per input
 - **Pulse stuffing** – add dummy bits to inputs with lower rates
- **Framing bits** follow a pattern, frame to frame, that allows the demultiplexer to synchronize with the incoming stream so that it can separate the time slots accurately

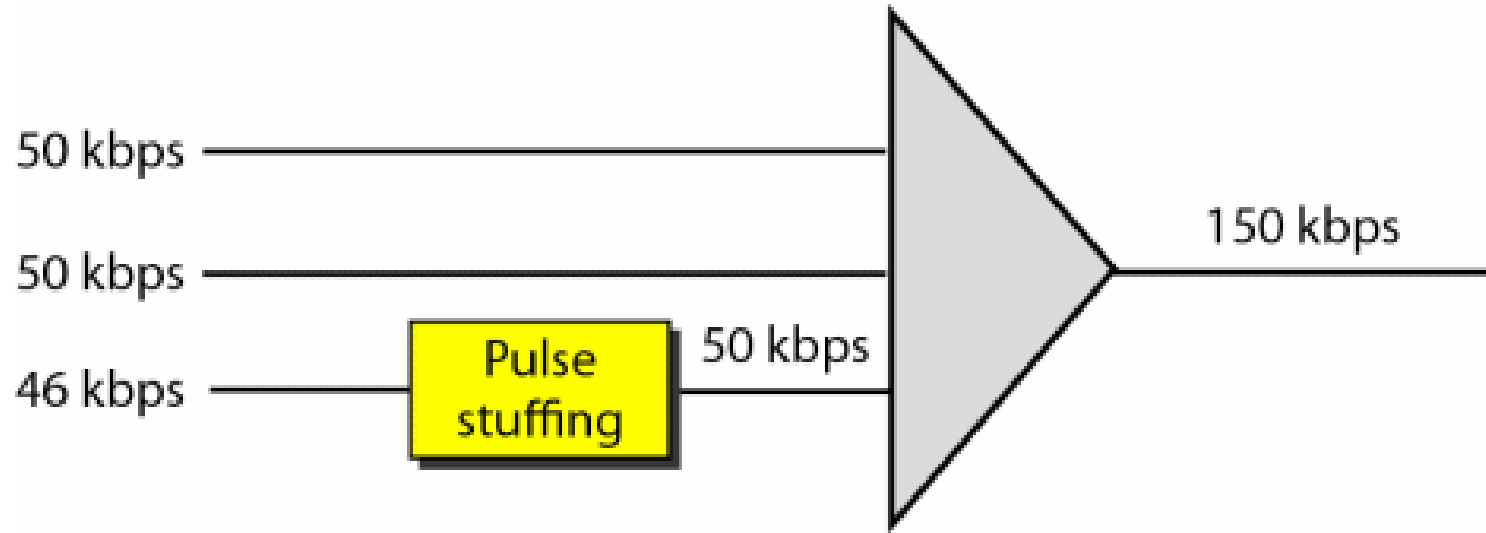
Multilevel Multiplexing



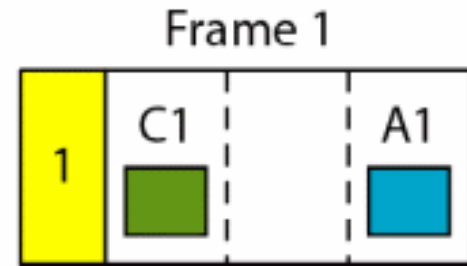
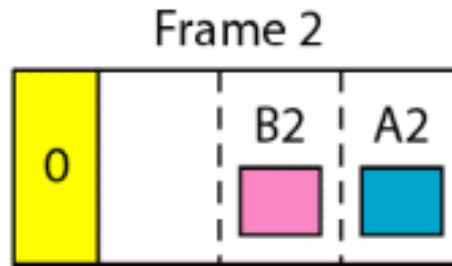
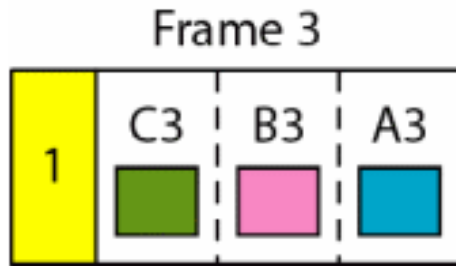
Multiple-slot Allocation



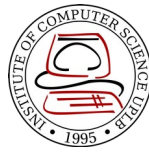
Pulse Stuffing



Framing bits

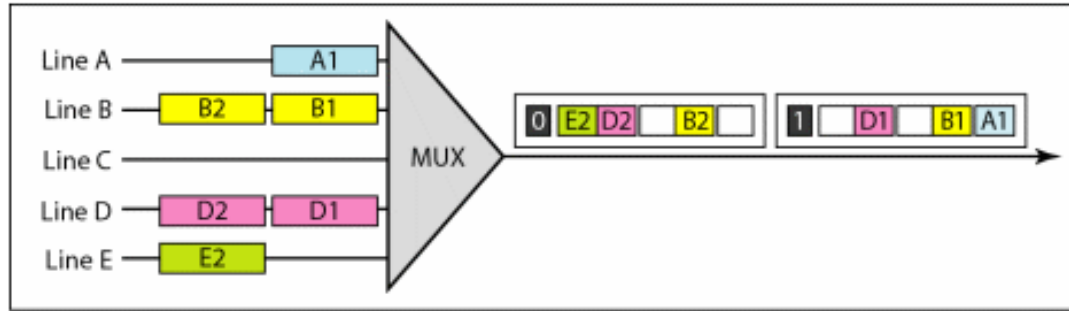
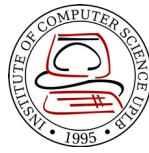


Statistical TDM

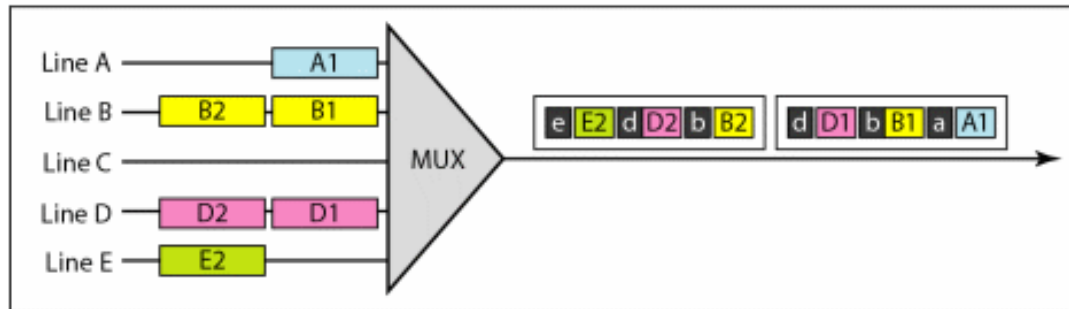


- Slots are dynamically allocated to improve bandwidth efficiency
- Number of slots in each frame is less than the number of input lines
- Address field is needed in each frame
- Data size-to-address size ratio must be reasonable
- No synchronization bits
- The capacity of the link is normally less than the sum of the capacities of each channel

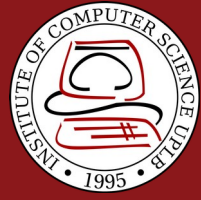
Synchronous vs Statistical TDM



a. Synchronous TDM

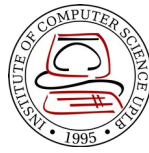


b. Statistical TDM

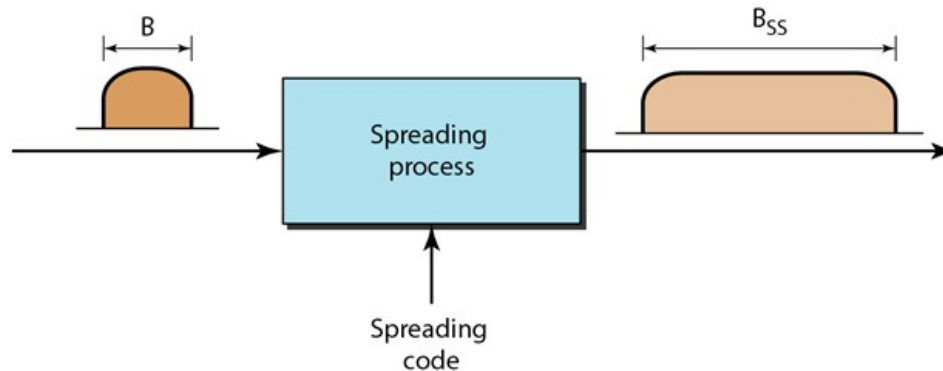


Spread Spectrum

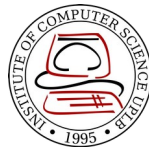
Spread Spectrum



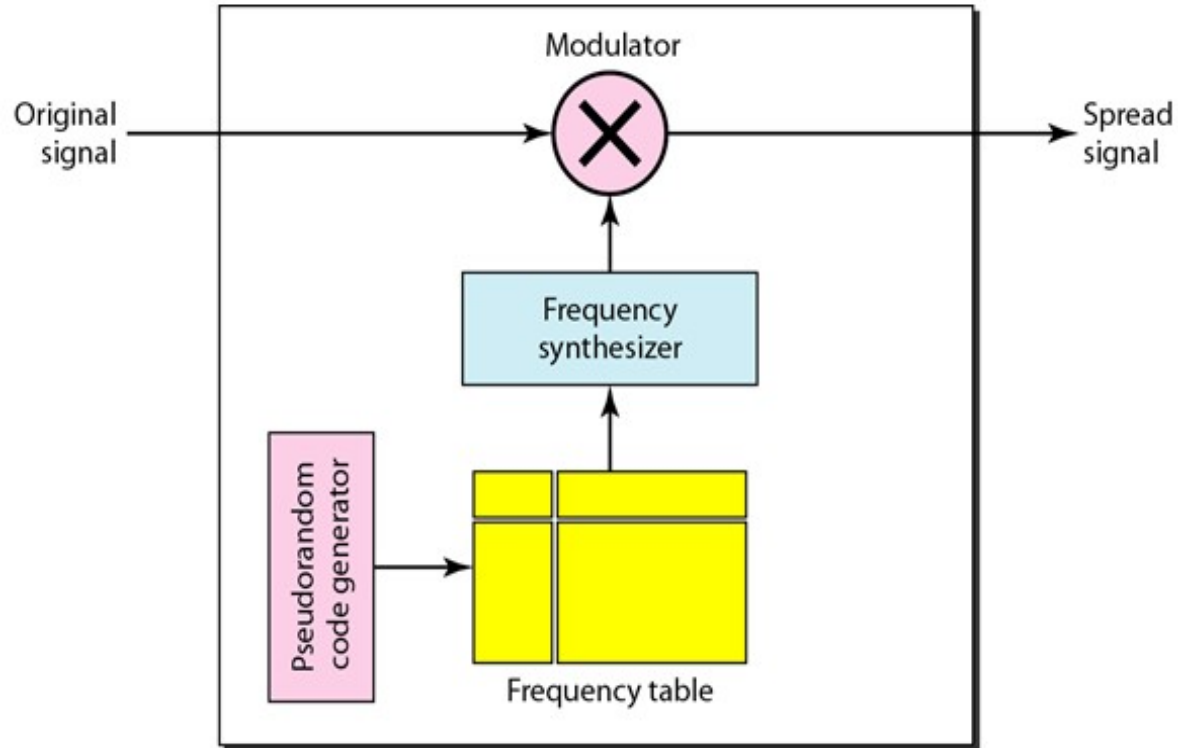
- For wireless applications
- Must protect from eavesdroppers and jammers
- Main technique is to add redundancy: if required bandwidth is B , $B_{ss} > B$
- Signals are spread with different codes so that they can be separated at the receivers
- Spreading process must occur after the signal is created by the source



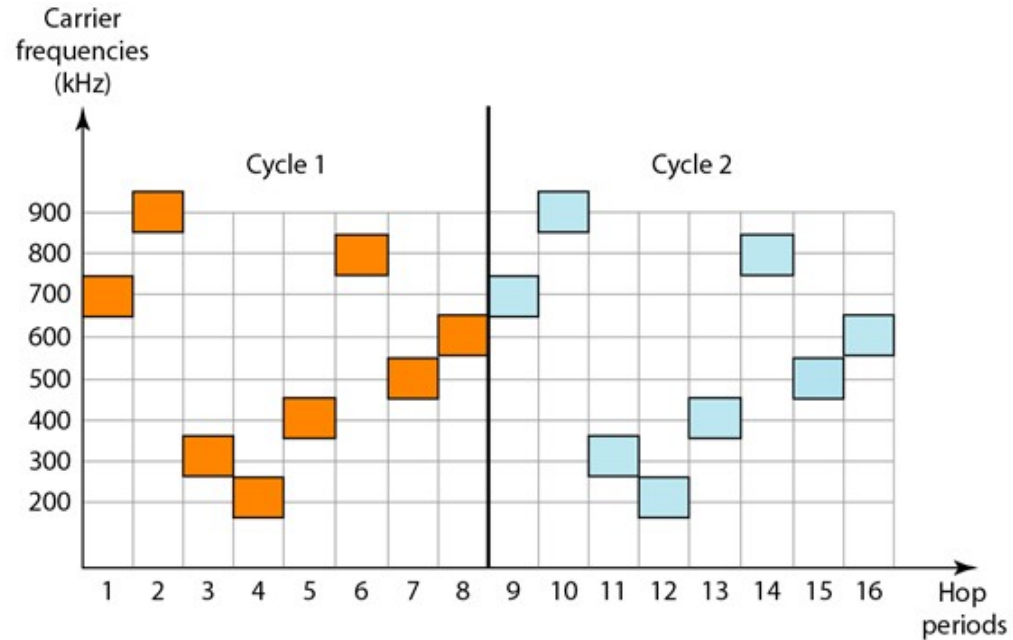
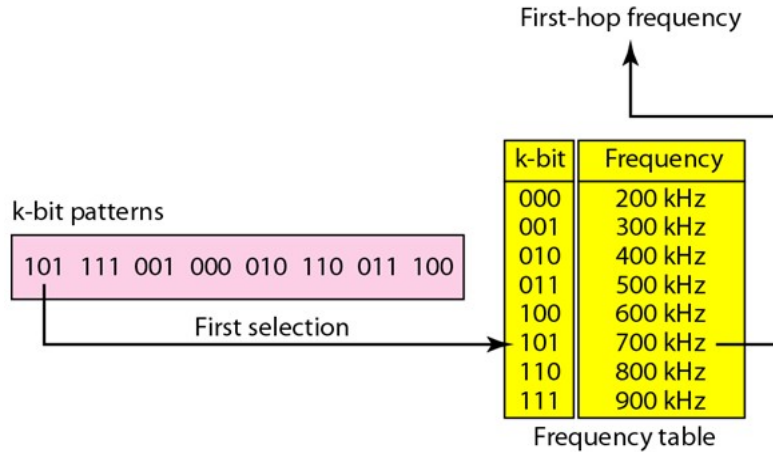
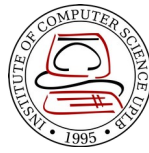
Frequency Hopping Spread Spectrum (FHSS)



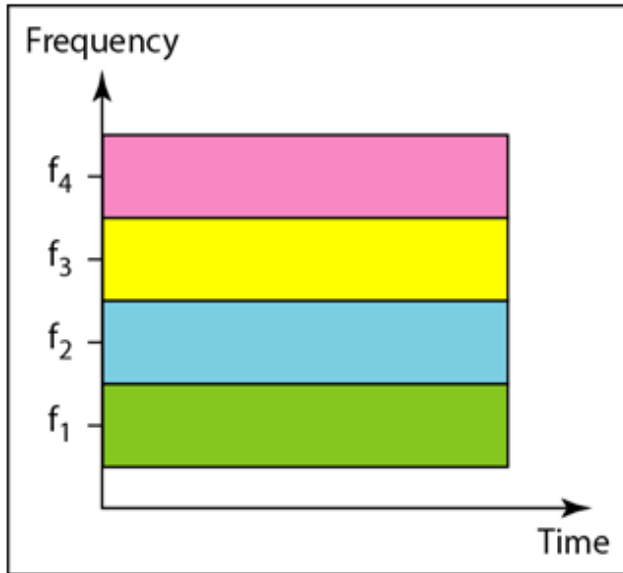
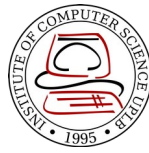
- Uses M carrier frequencies modulated by the source signal
- Modulation happens one carrier frequency after another
- Pseudorandom noise (PN) creates a k -bit pattern for every hopping period T_h
- The frequency table uses the pattern to find the frequency to be used for a hopping period and passes it to the frequency synthesizer
- Frequency synthesizer creates the carrier signal of the selected frequency



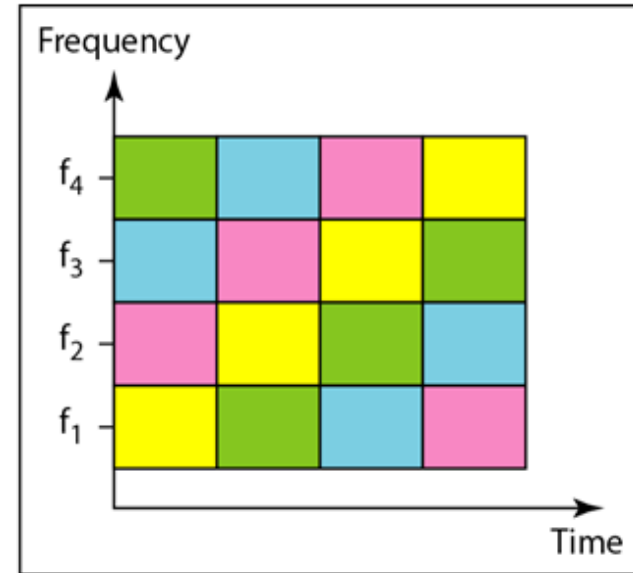
FHSS



FDM vs FHSS

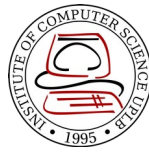


a. FDM

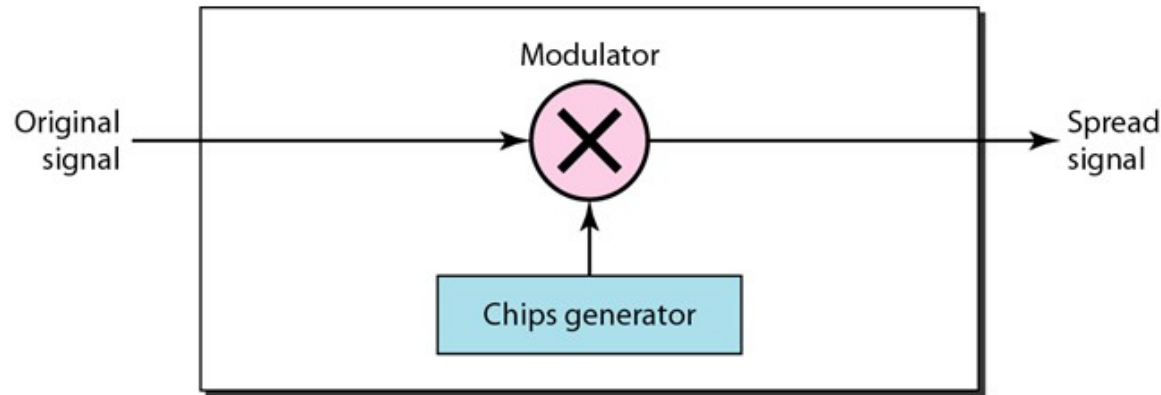


b. FHSS

Direct Sequence Spread Spectrum (DSSS)



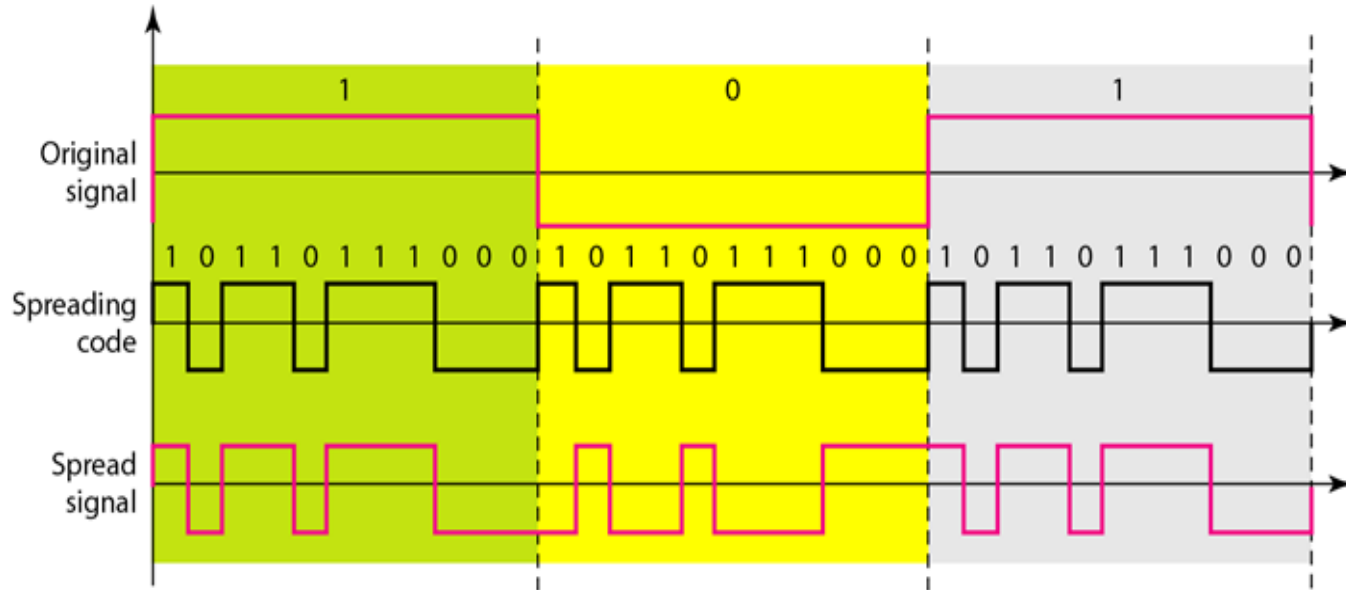
- Replace each data bit with n bits using a spreading code, called **chips**
- Chip rate is n times that of the data bit



DSSS



Barker sequence, where n is 11 bits



References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.